Ancestor relevance gate

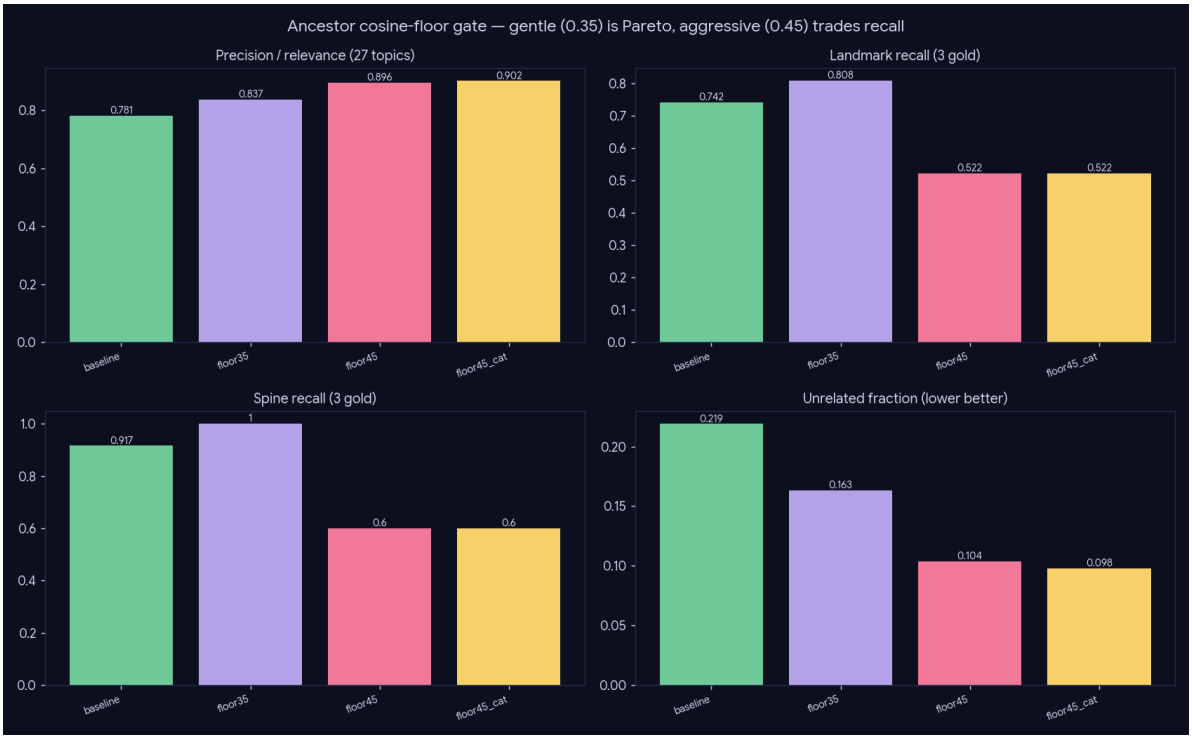
arxivkg · retrieval · plugging the noise mechanism

What we did

`02-02` found ∼22% of landmarks off-topic AND carrying high in-degree (5.4) — famous cross-field hubs (GPT-3 / StyleGAN under "diffusion"). They enter through one mechanism: `augment_with_ancestors` → `store.ancestors` pulls the top-120 *most-cited* one-hop-back papers (ranked by citer count *before* cosine), so fame beats relevance. We gate that injection by a cosine floor and measure **both** sides: **precision** (LLM `relevance_rate` over 27 topics, reusing the `02-02` cache) and **recall** (`landmark_recall` + `spine_recall` over 3 gold topics). Only the injected ancestors are gated; the core pool and cosine-gated `frontier_pull` are untouched.

Results

variant	precision (27)	unrelated	landmark rec. (3)	spine rec. (3)
baseline	0.781	0.219	0.742	0.917
floor35	0.837	0.163	0.808	1.0
floor45	0.896	0.104	0.522	0.6
floor45_cat	0.902	0.098	0.522	0.6



Conclusion

The tradeoff is **floor-dependent**. **A gentle floor (0.35) is a Pareto win, not a tradeoff**: precision 0.781 → 0.837 (robust, n= 27) *and* gold recall up (0.742 → 0.808, spine → 1.0). Removing clearly-off-topic ancestors (cosine < 0.35) frees PPR's 40 landmark slots for on-topic papers instead of famous hubs — both metrics rise together. **An aggressive floor (0.45) is the real tradeoff**: precision climbs to 0.896 but gold recall collapses to 0.522 (it cuts genuine low-cosine foundational papers — VAE / Sohl-Dickstein sit ∼0.4–0.5). The category gate adds ∼nothing (0.896 → 0.902); drop it. **Verdict:** a `min_cosine`≈0.35 floor on `augment_with_ancestors` is worth promoting to production — a low-risk Pareto improvement. **Caveat:** recall is n= 3 gold, precision n= 27; a wider gold set would harden the no-recall-loss claim.